

PRODUCTION, OPERATIONS, AND SUPPLY CHAIN MANAGEMENT

Somewhere between an idea and a customer's hands, a product has to actually get made and delivered. Today we trace that path — from the factory floor to the supply chain to the quality check at the end.

WHAT YOU'RE GETTING FOR YOUR 50 MINUTES

- 11.1** Describe the importance of operations management.
- 11.2** Explain how supply chain management creates and delivers products efficiently.
- 11.3** Explain how manufacturing and production are important to economies.
- 11.4** Discuss production processes and technology.
- 11.5** Describe production control and quality standards.

Gold-outlined boxes are in-class activities — that's your cue to actually participate, not just read.

THE DEFINITION WE BUILD ON ALL CLASS

Operations management. Planning and controlling the processes that turn resources into finished goods and services, with a focus on efficiency and profitability.

Chapter 10 covered the technology businesses use. Chapter 11 covers the physical work of actually making and moving things.

OBJECTIVE 11.1

Why operations management matters

Every product a customer touches passed through a series of decisions someone had to make on purpose.

Operations management. Planning and controlling production processes, with a focus on efficiency and profitability.

YOUR OWN EXAMPLE

Key decisions operations managers make

- Facility location — where to build or operate
- Capacity planning — how much the operation can produce
- Facility layout — how the physical space is organized for efficiency

REAL-WORLD EXAMPLE

Burger King built its whole operational model around “Have It Your Way” — a facility layout and process designed for customization at speed, not just standardized output.

NOTES · 11.1 WHY OPERATIONS MANAGEMENT MATTERS

OBJECTIVE 11.2

The supply chain behind every product

No company makes everything itself — every product depends on a network of other businesses working in sync.

Supply chain. The network of organizations involved in producing and delivering a product, from raw materials to the final customer.

YOUR OWN EXAMPLE

Supply chain management. Coordinating suppliers, manufacturers, distributors, and retailers so products flow efficiently.

YOUR OWN EXAMPLE

Important concepts

01 Make-or-buy decisions

Deciding whether to produce a component in-house or purchase it from an outside supplier.

EX.

02 Purchasing

Selecting suppliers and buying the materials a company needs to produce its goods.

EX.

03 Green supply chain management

Reducing the environmental impact of sourcing, production, and delivery.

EX.

IN-CLASS DISCUSSION

Should a company make a component itself or buy it from a supplier? What factors would tip that decision one way or the other?

NOTES · 11.2 THE SUPPLY CHAIN BEHIND EVERY PRODUCT

OBJECTIVE 11.3

Manufacturing drives economies

Manufacturing isn't just about the product — it's a major engine of local and national economic activity.

Two related, not identical, ideas

Production. The broad set of activities that create goods and services of any kind.

YOUR OWN EXAMPLE

Manufacturing. Specifically converting raw materials into finished physical goods.

YOUR OWN EXAMPLE

Why it matters

- Supports GDP
- Creates jobs
- Drives economic growth

REAL-WORLD EXAMPLE

Manufacturing plants like Sony DADC and Bemis have long supported jobs right here in Terre Haute — production drives local economies the same way it drives national ones.

NOTES · 11.3 MANUFACTURING DRIVES ECONOMIES

OBJECTIVE 11.4

How things get made

Not every product is made the same way — the right production system depends on volume and variety.

Five production systems

Tap each to see how it fits.

01 Continuous production

Runs nonstop, producing a standardized product around the clock (e.g., oil refining).

EX. _____

02 Job production

Creates a single, custom product to a specific order (e.g., a custom yacht).

EX. _____

03 Batch production

Produces goods in limited groups or batches (e.g., a bakery's daily bread run).

EX. _____

04 Mass production

Produces large volumes of a standardized product (e.g., a car assembly line).

EX. _____

05 Mass customization

Produces large volumes while still letting customers personalize the product (e.g., build-your-own sneakers).

EX. _____

Production technology

01 CAD

Computer-aided design — digitally designing a product before it's built.

EX. _____

02 CAM

Computer-aided manufacturing — using computers to control production equipment.

EX. _____

03 CIM

Computer-integrated manufacturing — linking design, manufacturing, and inventory systems together.

EX. _____

04 Robotics

Machines that perform physical production tasks automatically.

EX. _____

05 FMS

Flexible Manufacturing Systems — equipment that can be quickly reconfigured for different products.

EX. _____

DISCUSSION QUESTION

Think of a product that's mass-customized, like a build-your-own sneaker or a custom laptop. How is that different from pure mass production?

NOTES · 11.4 HOW THINGS GET MADE

OBJECTIVE 11.5

Keeping quality and output on track

Making the product is only half the job — a company also has to make sure it's making the right amount, at the right quality.

Production control. Scheduling and monitoring production to keep it running on time and on budget.

YOUR OWN EXAMPLE

Tools for staying in control

- Scheduling
- Inventory control
- ERP systems (Enterprise Resource Planning)
- MRP systems (Material Requirements Planning)

Quality management approaches

TQM. A company-wide commitment to continuously improving quality across every process.

YOUR OWN EXAMPLE

SPC. Using statistics to monitor and control a production process in real time.

YOUR OWN EXAMPLE

Six Sigma. A data-driven method for reducing defects to near-zero error rates.

YOUR OWN EXAMPLE

ISO Standards. International standards that certify a company's quality management systems meet a recognized benchmark.

YOUR OWN EXAMPLE

DISCUSSION QUESTION

Why might a company invest in ISO certification even though it's expensive and time-consuming?

NOTES · 11.5 KEEPING QUALITY AND OUTPUT ON TRACK

SPECIAL FOCUS

The make-or-buy decision

Almost every product involves this same question, over and over, at the component level: build it ourselves, or pay someone else to?

	Make	Buy
Control	High — full control over quality and timing	Lower — depends on the supplier
Capital needed	High — equipment, staff, facilities	Lower — pay per unit
Best for	Core, differentiating components	Standardized, widely available components

DISCUSSION QUESTION

Think of a product you know well. Which parts of it do you think the company makes, and which does it buy?

NOTES · THE MAKE-OR-BUY DECISION

CHAPTER 11 · WRAP-UP

Closing the assembly line

Define each term in your own words. If you can do all of them without looking, you're ready for the quiz.

DEFINE THESE IN YOUR OWN WORDS

- | | | |
|-------------------------|----------------------|---------------------------|
| • Operations management | • Supply chain | • Supply chain management |
| • Make-or-buy decision | • Manufacturing | • Mass production |
| • Mass customization | • CAD | • CAM |
| • Robotics | • Production control | • ERP |
| • MRP | • TQM | • Six Sigma |
| • ISO Standards | | |

ACTIVITY MINUTE · BEFORE YOU LEAVE

Which key term from this chapter matters most, and why?

Your one biggest “aha” moment from today.

Would you rather work in a make-focused or a buy-focused company, and why?

TOTAL — 50 MIN · PAID IN FULL

Chapter 10 covered the technology businesses use. Chapter 11 covered how products actually get made and delivered. Next, we look at how those products get in front of the right customer in the first place.